

Maintenance Operations Guide

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1. Preventive Maintenance Schedule

All production equipment follows a tiered preventive maintenance (PM) schedule based on criticality classification and manufacturer recommendations.

Criticality	PM Frequency	Inspection Type	Max Downtime
Critical	Weekly	Full diagnostic	4 hours
High	Bi-weekly	Visual + functional	8 hours
Medium	Monthly	Visual inspection	24 hours
Low	Quarterly	Basic check	48 hours

Overdue Maintenance

Equipment is flagged as overdue when `NEXT_SERVICE_DATE < CURRENT_DATE`. Critical equipment overdue triggers an immediate work order and production line review.

2. Work Order Management

All maintenance activities must be documented through the Work Order Management System (WOMS). No maintenance action may be performed without an active work order.

Priority	Response Time	Escalation	Approval Required
Emergency	< 1 hour	Immediate to Ops Mgr	None (post-facto)
Urgent	< 4 hours	Shift Supervisor	Supervisor
Planned	Scheduled	N/A	Maintenance Lead
Deferred	Next PM window	Weekly review	Plant Manager

Work orders must include: equipment ID, fault description, parts required, estimated duration, safety lockout/tagout confirmation, and assigned technician.

3. Equipment Inspection Checklist

Technicians must complete the following inspection points during each preventive maintenance visit:

Category	Check Item	Pass Criteria
Electrical	Motor amperage	Within +/- 10% of nameplate
Electrical	Insulation resistance	> 1 M-ohm
Mechanical	Vibration levels	< 4.5 mm/s RMS
Mechanical	Belt/chain tension	Per manufacturer spec
Lubrication	Oil level and quality	Within operating range

Safety	Guard integrity	All guards in place
Safety	E-stop function	Responds within 0.5s

Any failed inspection point requires immediate documentation in the work order and escalation based on equipment criticality level.

4. Downtime Tracking and Analysis

All equipment downtime events must be classified and recorded for reliability analysis.

Downtime Type	Definition	KPI Impact
Planned	Scheduled PM or upgrade	Planned Downtime %
Unplanned - Breakdown	Unexpected equipment failure	MTBF, MTTR
Unplanned - Quality	Stopped for quality issue	OEE, Defect Rate
External	Utility, supply chain	Excluded from OEE

Reliability KPIs

MTBF (Mean Time Between Failures) target: > 720 hours for critical equipment. MTTR (Mean Time to Repair) target: < 4 hours for critical, < 8 hours for high priority.

Root cause analysis (5-Why or Fishbone) is required for any unplanned downtime event exceeding 2 hours on critical equipment.

5. Energy Management and Budgeting

Each facility operates under an annual energy budget. Monthly consumption is tracked against budget with automated variance alerts.

Energy Type	Budget Metric	Variance Alert	Escalation
Electric	kWh per month	> 20% over budget	Facility Manager
Natural Gas	Therms per month	> 15% over budget	Facility Manager
Compressed Air	CFM baseline	> 10% over baseline	Maintenance Lead

- Peak demand monitoring: PEAK_DEMAND_KW tracked per meter to identify demand charge optimization opportunities.
- Cost allocation: Energy costs allocated to production lines based on sub-metering where available.
- Efficiency projects: Any project with payback period < 3 years is automatically approved for capital expenditure.
- Reporting: Monthly energy dashboard reviewed by Plant Manager with quarterly executive summary.